

Lab 7

April 7, 2022

1 Finding Transfer Function

FIR: Given - Zeros: $(-0.7, \pm 0.1j)$; Poles: $(0, 0)$ Calculation:

$$Zeros = (z - (-0.7 + 0.1j)) * (z - (-0.7 - 0.1j))$$

$$Poles = (z - 0) * (z - 0)$$

$$H(z) = \frac{z^2 + 1.4z + 0.5}{z^2}$$

$$\frac{Y(z)}{X(z)} = \frac{z^2 + 1.4z + 0.5}{z^2}$$

$$\frac{Y(z)}{X(z)} = \frac{z^{-2} z^2 + 1.4z + 0.5}{z^{-2} z^2}$$

$$\frac{Y(z)}{X(z)} = 1 + 1.4z^{-1} + 0.5z^{-2}$$

Then find the difference equation $Y[n] = X[n] + 1.4X[n-1] + 0.5X[n-2]$

IIR: Do the same calculation for IIR and find the difference equation. The difference equation in IIR may contain $Y[n-1]$, .. at the right hand side.

2 Sample Example

Say, waveform generator is producing a noise signal ($500kHz$, $500mV$ amplitude, and $500mV$ offset) from one channel and a sinusoidal signal ($1kHz$, $1V$ amplitude, and $1V$ offset) from another channel. This means that the signal on the second channel will oscillate between $0V$ and $2V$ at a frequency of $1kHz$. The difference equation is given: $Y[n] = X[n] + \sqrt{2}X[n-1] + X[n-2]$

Code Sample: FIR

```
input = adcddata1 + adcddata2 - 2048
ip[0] = input/8;
dacdata1 = ip[0] + sqrt(2)*ip[1] + ip[2] + 2048
ip[2] = ip[1]
ip[1] = ip[0]
dacdata1 = adcddata1 + adcddata2
```

3 Visualizing Output

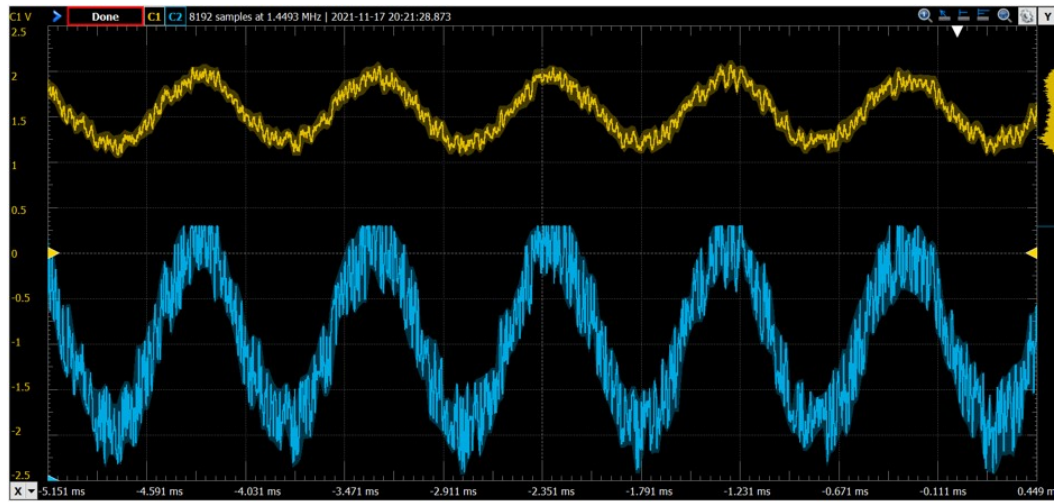


Figure 1: Using an FIR filter to clean the sine wave